

Paper B: corrections to the manuscript (master plan)

Against the author’s draft committed at e7b997e3, 30 September 2026

Each row sets the current manuscript text (left) against the proposed replacement (right), both typeset as they would appear. BEFORE text is cut from the committed manuscript by script, so it matches the manuscript exactly. Under each entry, **Why** gives the error and **Evidence** the Lean or sympy record or page that settles it (audit items refer to `notes/citation_audit.md`). Entry C.15 (inside C.11.9) reconciles Section 3 with `notes/positioning_economics.tex`; the abstract (C.1) leads with the identification point for the same reason. Cross-references print as their label (e.g. PROP:IMM); footnotes print inline. **Nothing here is applied.** The author approves entries by number; approved entries are applied to the manuscript, compiled and committed. Every AFTER text passes the mechanical checks of `notes/writing_discipline.md` (no colons, no dashes doing a sentence’s work, “sequence” for reading order, the fixed adoption terminology, length within 80–120% of the draft except for deletions, citation fixes and the removed duplicate); the generator refuses to build otherwise.

Section C: corrections to the current draft

Part	BEFORE (current manuscript)	AFTER (proposed)
C.1	Abstract, sentences 2-4	
	Considering a case where evaluators reading two correlated soft cues compare their beliefs as well as decisions against a sequence-free Bayesian benchmark, the current papers asks which of the beliefs and decision statistics register the resulting order effect. Considering the empirical limitations in addition to theoretical underpinnings, the paper shows that an arbitrary statistic is an order of magnitude closer to the benchmark when two cue-sequences displace a belief within a two-dimensional plane. This results in measures of the believed association and the average value lost through decisions differing from benchmark staying close to the benchmark while the share of the population affected by the sequence differing from the benchmark.	Considering evaluators who read two correlated soft cues and comparing their beliefs and decisions with a sequence-free Bayesian benchmark, the paper asks which statistics of those beliefs and decisions register the resulting sequence effect. Whether a statistic registers it is a question of identification rather than of estimation, since the two reading sequences move a belief in only two directions, and a statistic insensitive to both carries no first-order signal in the population itself, however many evaluators are sampled. The believed association between the attributes and the average value lost through decisions that differ from the benchmark are of that kind, while the marginal probabilities and the share of the population whose decision the sequence changes depart from the benchmark at first order.
Why. “An arbitrary statistic is an order of magnitude closer to the benchmark” is false as stated; only the protected statistics are (Proposition PRO; plan 0.A made the same point). The two-dimensional plane is the mechanism, not the finding; for the Journal of Mathematical Economics the abstract leads with the economic content, that a null on a belief audit is an identification failure and not an estimation one (the point of <code>notes/positioning_economics.tex</code> , already in the conclusion at MS ~950). Also “the current papers asks”, and “This results in measures ... staying ... while the share ... differing” has no main verb. “Considering the empirical limitations in addition to theoretical underpinnings” is cut because no empirical limitation is named.		
Evidence. DEC and LOS (second order), DRF and SHR (first order), PRO (the plane and the classification). All in the Lean and sympy suites.		
C.2	Introduction, paragraph 1, sentences 2 and 4	
1/2	The sequence-dependence of Jeffrey conditioning for soft evidence has attracted much debate (Diaconis and Zabell, 1982; Hawthorne, 2004) but the mechanism of how sequence dependence register in observed statistics remains unclear (Zhao and Osherson, 2010).	The sequence dependence of Jeffrey conditioning on soft evidence has attracted debate (Diaconis and Zabell, 1982; Hawthorne, 2004; Zhao and Osherson, 2010), but how that dependence registers in the statistics an observer reads has received little attention.

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2/2	It finds that the share of population affected by evidence sequence deviates from a sequence-invariant benchmark – but the believed association between attributes and the total loss due to sequence effects do not.	It finds that the share of the population whose decision the sequence changes departs from a sequence-invariant benchmark at first order, while the believed association between attributes and the surplus-weighted loss depart from it only at second order.
Why. Zhao-Osherson do not say the mechanism “remains unclear”; they say doubts about order dependence “disappear on closer inspection of the evidential weight of probability judgements” (p. 291). Cited for the debate, as the author agreed. Jeffrey himself says non-commutativity “is as it should be” (1983, pp. 182-183; audit J1) and Diaconis-Zabell that it “is not a real problem” (Remark 2, p. 827; M24); Hawthorne calls the order effect “very troubling” (p. 99). Sentence 4: the loss is not free of the sequence effect, it carries it at second order (LOS), and “– but” is a dash doing a sentence’s work. Evidence. verify_jeffrey.md J1; verify_kinematics.md; verify_hawthorne_weisberg.md; ZO p. 291 read 2026-09-30; Theorem LOS, Proposition SHR.		
C.3 Introduction, paragraph 2 (the literature paragraph) and its footnote		
	The discussion on Jeffrey (Jeffrey, 1983) conditioning’s dependence on arrival sequence have focused more often on whether the commutation criterion is exactly satisfied or not but not on how measurable it is in observed data. The impression literature (Asch, 1946) in fact often models an impression as a single score, one attribute at a time, so it has only one statistic to read. While the characterisation of how two sequences could lead to the same belief Diaconis and Zabell (1982); Hawthorne (2004) does consider a joint belief over attributes, the order dependence is more likely to be treated as a defect to repair <i>[Footnote: Hawthorne (2004) repairs this by letting a cue multiply the old belief instead of replacing it, and under his order-free variants the sequence effect disappears.]</i> rather than as something to measure. While the changing of input from a replacing credence to a multiplying factor settles whether beliefs depend on sequence or not, it hardly addresses the empirical issue that an observer cannot tell whether impressions replace prior belief or adjust it.	Discussion of how Jeffrey conditioning (Jeffrey, 1983) depends on arrival sequence has more often asked whether the commutation criterion is exactly satisfied than how measurable a failure of it is in observed data. The belief-adjustment literature models an impression as a single score (Hogarth and Einhorn, 1992), and the founding multi-trait study reports its results trait by trait (Asch, 1946), so both read marginals only. The characterisation of when two sequences lead to the same belief does consider a joint belief over attributes (Diaconis and Zabell, 1982), but where sequence dependence has been treated as a problem it has been treated as a defect to repair <i>[Footnote: Hawthorne (2004) offers factor-based alternatives in which a cue supplies a normed-likelihood or likelihood-ratio factor instead of a new probability. Under these, updates on distinct attributes commute, and in his Basis-Commuting Version all updates do.]</i> rather than as something to measure. Replacing a credence that resets the prior marginal by a factor that multiplies it settles whether beliefs depend on sequence, but it leaves open the empirical question of whether an impression replaces a prior belief or adjusts it.
Why. (i) Asch did not model an impression as a single score; Experiment VI reports eighteen traits under two orders (audit A28). His published tables give per-trait percentages, i.e. marginals. (ii) Diaconis-Zabell are grouped with the defect view, but they hold that non-commutativity “is not a real problem” (M24). (iii) The footnote says Hawthorne “repairs” the defect and that the effect disappears; he offers alternatives, and order-freedom holds across distinct bases, fully only in his Basis-Commuting Version (M19). (iv) \citet inside a sentence where \citep is meant; “have focused” agreement. Evidence. Asch1946 record (Asch.lean, Table 7); DiaconisZabell.lean and README; Hawthorne.lean (<code>extUpdate_basisCommuting</code>, <code>factorUpdate_comm</code>), Sections 6-8.		
C.4 Introduction, the two-channel paragraphs (appear twice)		

Part	BEFORE (current manuscript)	AFTER (proposed)
	More specifically, arrival-sequence dependence can enter through two channels, an as- sociation channel, in which one cue changes what the other implies and which exists only when the attributes are believed related, and a position channel, in which the observer weights the later cue less whatever the attributes are (Hogarth and Einhorn, 1992; Asch, 1946). Unlike the position channel, where the later cue is adopted only in part and every marginal registers the sequence even when the attributes are believed unrelated, the association channel is one where the sequence registers unevenly across statistics due to each impression being adopted in full (Diaconis and Zabell, 1982) so that the position channel is absent. More specifically, arrival-sequence dependence can enter through two channels, an association channel, in which one cue changes what the other implies and which exists only when the attributes are believed related, and a position channel, in which the observer weights the later cue less whatever the attributes are (Hogarth and Einhorn, 1992; Asch, 1946). Which channels operate is fixed by how fully the later cue is adopted. When it is adopted only in part, the position channel operates and every marginal registers the sequence even when the attributes are believed unrelated. When each impression is adopted in full (Diaconis and Zabell, 1982), the position channel is absent and the association channel is the only one – so that the sequence registers unevenly across statistics. The association channel is the setting that the current paper explores. In the two-attribute setting, the paper therefore considers three coordinates of a belief – two marginal probabilities and the cross-attribute association – and explains how a difference from order-free benchmark below the observer’s precision is invisible in the statistic read by the observer. The question that paper addresses is not simply whether the two sequences agree but by how much they disagree in an arbitrary statistic. The answer the paper finds and details is that whereas the marginal probabilities and decisions thereof carry the difference at first order in the prior covariance, the believed association and related statistics carry it only at second order.	More specifically, arrival-sequence dependence can enter through two channels. In the association channel one cue changes what the other implies, and the channel exists only when the attributes are believed related. In the position channel the weight a cue receives depends on where in the sequence it arrives, whatever the attributes are. When the first cue is adopted in full and later cues only in part the earlier impression is favoured, and when every cue is adopted only in part the later one is (Hogarth and Einhorn, 1992). In the model studied here, which channels operate is fixed by how fully the later cue is adopted. When it is adopted only in part, the position channel operates and every marginal registers the sequence even when the attributes are believed unrelated. When each impression is adopted in full, as in the successive updating of Diaconis and Zabell (1982), the position channel is absent and the association channel is the only one, so the sequence registers unevenly across statistics. That full-adoption setting is the one the current paper explores. A belief in the two-attribute setting has three coordinates, the two marginal probabilities and the cross-attribute association, and a departure from the sequence-free benchmark smaller than the observer’s precision is invisible in the statistic read. The question is therefore not only whether the two sequences agree but by how much they disagree in a given statistic. The marginal probabilities and the decisions made from them carry the difference at first order in the prior covariance, while the believed association and the statistics that move with it carry it only at second order.
Why. (i) The paragraph appears twice; the first copy is an earlier version and lacks its final period. The AFTER replaces both copies. (ii) “the observer weights the later cue less” is cited to Hogarth-Einhorn and Asch, but Hogarth-Einhorn’s step-by-step partial adjustment predicts recency (Appendix B) and their primacy comes from anchoring on the first item (End-of-Sequence, Eq. 8); Asch denies that “sheer temporal position” matters (p. 272). Asch is dropped (audit P10-P12). (iii) “Which channels operate is fixed by how fully the later cue is adopted” is true of this model but not in general (under base-rate neglect the later cue is adopted in full and a position channel still operates), so it is scoped to “the model studied here”. (iv) Full adoption is Diaconis-Zabell’s setting, not their premise. Optional, if a published instance of a position channel with likelihood inputs is wanted: add after the Hogarth-Einhorn sentence “Under base-rate neglect, where each posterior becomes the prior for the next cue, the earlier cue is discounted \citep{BenjaminBodohCreedRabin2019}.” (needs the bib entry in C.12). Evidence. HogarthEinhorn.lean (appB_recency, eq8_estimation_first_dominates, oneSided_primacy_iff); BenjaminBodohCreedRabin.lean; verify_interior_omega.py rows 1-2; Proposition ADJ endpoints (Anchoring.lean).		
C.5 Introduction, hiring-panel paragraph, last sentence		
	In demonstrating how the sequence-dependence of certain statistics can be invisible to an observer, the panel example also clarifies that the invisibility does not depend on the extent to which a later impression erases the earlier one (the amnesic updating concern in the literature).	In the panel example the odds ratio between the two traits is the same in both reading sequences however far a later impression erases the earlier one, since each update rescales rows or columns of the belief (Lemma LEM:SEP), whereas the cross-product association is protected in that way only under full adoption (Section SEC:SCOPE).
Why. False as written for the cross-product association, which is the paper’s assoc. Under partial adoption its sequence effect is first order in c, with coefficient $(1-\omega)$ times a nonzero term, and the pooled association is nonzero already at $c = 0$. Only the odds ratio is identical across sequences for every ω. The commented-out line below the paragraph states the odds-ratio version. The AFTER points to Section 6, where plan 6.B carries the interior-omega numbers; apply 6.B with it. Evidence. sympy/verify_interior_omega.py checks (2), (3), (4); Lemma SEP (LemmaSEP.lean, general N).		

Part	BEFORE (current manuscript)	AFTER (proposed)
C.6 Introduction, premise paragraph, from “Read as a Bayes factor”		
	Read as a Bayes factor, the credential carries a likelihood ratio stating how much more probable it is under competence than in its absence, and that ratio multiplies whatever belief it meets. A Bayes factor, however, requires the probability of the same credential for a candidate who is not competent, and an impression does not supply that counterfactual. As a mechanism that holds conditionals fixed, Jeffrey updating is the unique coherent revision for credence delivered on the cue’s partition and is equivalent to making the minimal change of the prior consistent with the delivered marginal (Diaconis and Zabell, 1982)	Read as a Bayes factor, the credential carries the ratio of new to old odds on competence, which multiplies whatever belief it meets. The two readings agree on a single cue (Proposition PROP:IMM) and differ in what stays fixed when the cue meets another prior. The panelist’s own impression is naturally a credence, whereas a factor suits evidence reported by someone else, whose report mixes the evidence with that person’s prior (Jeffrey, 2004, ch. 3). Jeffrey updating holds fixed the conditionals given the cue’s partition, and its posterior is the unique belief with the delivered marginal closest to the prior in Kullback–Leibler divergence (Diaconis and Zabell, 1982, Theorem 5.1)
Why. (i) “A Bayes factor requires the probability of the same credential for a candidate who is not competent” is contradicted by Jeffrey’s own definition, new odds over old odds, which needs no likelihood (2002 draft of Jeffrey 2004, ch. 3; audit J2). Jeffrey draws the line between the readings by provenance instead (own experience gives credences, others’ reports should be converted to factors), which supports the paper’s modelling choice. (ii) “the unique coherent revision” is not in Diaconis-Zabell; what they prove is the unique Kullback-Leibler (and Hellinger) minimiser (M22). “equivalent to making the minimal change” without naming the distance is also loose (not unique in variation distance). The footnote after the citation is unchanged. Needs the Jeffrey2004 bib entry (C.12); the Drive copy is the 2002 draft, so cite the chapter, not a page. Evidence. verify_jeffrey.md J2; DiaconisZabell.lean (thm51_KL_eq_iff, jcond_iff_jeffrey); PropIMM.lean (propIMM_single_cue).		
C.7 Introduction, terminology paragraph, Domotor sentence and two typos		
1/2	The sequence-dependence of marginals follows easily – since the likelihood a delivered credence implies must be read against the marginal in force when the cue arrives (Domotor, 1980).	The sequence dependence of the marginals follows directly, since a delivered credence is taken against the marginal in force when the cue arrives, which the first cue moves when $c \neq 0$ (Proposition PROP:DIV).
2/2	To avoid a clash in terminology we use “order” to describe proximity with a sequence-free Bayesian benchmark and “sequence” to denote the sequence in which evidence is read.	To avoid a clash in terminology we use “order” to describe proximity with a sequence-free Bayesian benchmark and “sequence” to denote the sequence in which evidence is read.
Why. Domotor never mentions likelihoods or marginals; he supports only bare non-commutativity, and treats it as a defect of Jeffrey machines (p. 395), the opposite lean (audit M1). The mechanism is the paper’s own (Proposition DIV). If a Domotor citation is wanted, it fits the literature paragraph as a source of the commutativity demand that Jeffrey (1983, pp. 182-183) rejects. Typos: “bechmark”, sequence closed with backquotes. Evidence. Domotor.lean (field_eq_jeffrey_embed, embed_depends_on_state); Cripps.lean (composite_AB_eq_bayes: the B-marginal moves by $c(q_0 - \alpha) / (\alpha(1 - \alpha))$).		
C.8 Introduction, key-finding paragraph, last two sentences		
	The implication for evaluation is that and audit’s answer depends on which statistic is aggregated rather than on how many evaluators are averaged over. One consequence, is that the instruments most naturally trusted for detecting stereotype—belief audits asking how attributes are thought to go together—may not even be detectable.	The implication for evaluation is that an audit’s answer depends on which statistic is aggregated rather than on how many evaluators are averaged over. One consequence is that belief audits asking how attributes are thought to go together, the instruments most naturally trusted for detecting stereotype, cannot detect the sequence effect at first order.
Why. Typos (“and audit’s”, “One consequence, is”), and “the instruments ... may not even be detectable” says the instruments are undetectable; it is the sequence effect that belief audits cannot detect at first order. Evidence. Propositions DEC, PRO.		
C.9 Setup 2.1, the prior paragraph and the sequence/weight sentences		
1/3	The three numbers that fix P are described as Δ^3 – specified using the marginals $P(A=0) = \alpha$, $P(B=0) = \beta$ and the prior covariance c between the attributes:	The prior is a point of Δ^3 fixed by three numbers, the marginals $P(A=0) = \alpha$ and $P(B=0) = \beta$ and the prior covariance c between the attributes,
2/3	Simply put, the marginals $\alpha = P(A=0)$ and $\beta = P(B=0)$ denotes the shares of applicants the panelist believes to lack competence and to lack trustworthiness and the prior covariance $c = \text{assoc}(P)$ denotes how strongly the panelist believes the	Simply put, the marginals $\alpha = P(A=0)$ and $\beta = P(B=0)$ denote the shares of applicants the panelist believes to lack competence and to lack trustworthiness, and the prior covariance $c = \text{assoc}(P)$ denotes how strongly the panelist believes the

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3/3	To address the issues in amnesic updating debate, we also consider an adoption weight $\omega \in [0, 1]$ – which denotes how far the letter displaces what the credential had already implied about trustworthiness. At $\omega = 1$ the letter sets the rating outright, and at $\omega = 0$ the credential’s implication is never moved.	To address the debate over whether a later impression should be adopted in full (Hawthorne, 2004), we also consider an adoption weight $\omega \in [0, 1]$, which denotes how far the letter displaces what the credential had already implied about trustworthiness. At $\omega = 1$ the letter sets the rating outright, and at $\omega = 0$ the credential’s implication is never moved.
Why. Grammar ("described as Δ^3 –“, "denotes“ for two subjects). "Amnesic" is used only where Hawthorne is quoted (writing_discipline.md 6). Dependency: omega is defined here but used nowhere else in the manuscript until plan 5.B (Proposition ADJ) is applied; either apply 5.B or move this sentence into 5.B. The Frechet-Hoeffding bounds in the new text are correct (checked cell by cell). Evidence. Bounds: nonnegativity of the four cells of P; Hawthorne pp. 98-99.		
C.10 Setup 2.2, the invariance condition and the justification of the rule		
1/2	holding the conditionals fixed—the invariance condition (Jeffrey, 1983, ch. 11).	holding the conditionals given the cue’s partition fixed, the invariance condition (Jeffrey, 2004, ch. 3), which Jeffrey (1983, ch. 11) expresses by saying that the change originates in the partition.
2/2	The axiomatic grounding is twofold. The partition satisfying the invariance condition is the minimal sufficient statistic for revising the prior to any candidate posterior on that partition (Diaconis and Zabell, 1982), and the rule is the minimal revision of the prior consistent with the delivered marginal, in the I -divergence sense (Diaconis and Zabell, 1982).	Two results of Diaconis and Zabell (1982) support the rule. The invariance condition makes the cue’s partition sufficient for the prior and posterior, and minimal whenever the delivered credence differs from the prior marginal (their Theorem 2.2). The rule is also the unique minimiser of the I -divergence from the prior among all beliefs with the delivered marginal (their Theorem 5.1).
Why. (i) Chapter 11 of Jeffrey (1983) never uses “invariance”; it says the change “originates in” the partition (pp. 168, 174). “Invariance” is the 2004 book’s term (audit J4). (ii) “The partition satisfying the invariance condition is the minimal sufficient statistic for revising the prior to any candidate posterior” misstates D-Z Theorem 2.2: sufficiency is for one pair $\{P, P^*\}$, every refinement of such a partition also qualifies, and the minimal one is the likelihood-ratio partition (M23; Jeffrey p. 174 also allows “a certain latitude in the choice” of partition, J3). (iii) D-Z give no axioms; they call this “mechanical updating”. The summation index is x because ω is now the adoption weight. Evidence. DiaconisZabell.lean (thm22_lr_minimal, jcond_of_refines, lr_jeffrey_iff, thm51_KL_eq_iff); verify_jeffrey.md J3-J4.		
C.11 Related literature, the paragraph from BHW/Banerjee to Becker and Good-Mittal		
1/9	In Banerjee (1992), the sequence persists because a coarse public action fails to be a sufficient statistic for private information. In Bikhchandani et al. (1992), on the other hand, a cascade is an action that conveys no private signal.	In both, the equilibrium action is not a sufficient statistic for private information, being binary in Bikhchandani et al. (1992) and a non-invertible choice rule in Banerjee (1992), so once a cascade or herd forms, later actions convey no private signal.
2/9	One similarity with the literature, however, is that the sequence effect in (Bikhchandani et al., 1992; Banerjee, 1992) lives on the analogue of the marginals—which remain so in the current paper as well.	One analogy remains. Those models have a single state, so their sequence effect falls on a belief about one variable, the counterpart of a marginal, and here too the marginals carry the sequence effect.
3/9	A related normative literature evaluates group belief formation by whether aggregation commutes with updating—a design criterion for the pooling rule (Dietrich, 2021). While geometric pooling (Dietrich, 2021, Def. 1) satisfies the criterion, the population mean studied in the current paper (i.e. the average posterior across evaluators who differ only in the sequence they meet the two cues) is the outcome of linear pooling and generically fails the criterion. As further sections elaborate, this is consistent with the first-order divergence of every statistic outside the protected class of Proposition PROP:PRO.	A related normative literature evaluates group belief formation by whether aggregation commutes with updating, a criterion that geometric pooling meets and linear pooling generally does not (Dietrich, 2021, Theorem 2). The criterion does not discriminate here. Evaluators who share a prior and differ only in reading sequence do not hold conditionalisations of that prior on a common event, and the linear and geometric averages of their posteriors differ only at $\mathcal{O}(c^2)$. The first-order divergence of every statistic outside the protected class of Proposition PROP:PRO is therefore a property of Jeffrey updating, not of linear pooling.

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4/9	Several alternatives to Bayesian updating have been used in the decision theory literature to explain whether sequential conditioning is sequence-independent or not. For example, Epstein (2006) makes the updating rule subjective, and Ortoleva (2012) axiomatises departures triggered by unexpected news. Cripps (2021) shows that symmetry and divisibility jointly force sequence-independence, so the correlated two-cue composite of Proposition PROP:DIV fails divisibility—and, of the four axioms, only divisibility. <i>[Footnote: Cripps’s axioms govern rules on fixed-likelihood experiments; the statement here is behavioural—the map from prior and delivered credences to final belief, held against the sequence-invariance that his Symmetry and Divisibility axioms jointly force.]</i>	Decision theory has axiomatised several alternatives to Bayesian updating, each for a single piece of news. Epstein (2006) makes the updating rule, and not only the prior, subjective, and Ortoleva (2012) axiomatises departures triggered by unexpected news. Cripps (2021) treats the sequence of signals, and his Symmetry and Divisibility axioms imply that two conditionally independent signals give the same posterior in either sequence. Read with each cue as its likelihood matched to the prior (Proposition PROP:IMM), the composite of Proposition PROP:DIV is a sequence of Bayes updates satisfying all four of his axioms, and it depends on the sequence because the second likelihood is matched to the intermediate belief, so the two sequences process different experiments.
5/9	While the literature restores sequence-invariance for sequential Jeffrey updating by changing how successive inputs are pooled (Pettigrew and Weisberg, 2025),	While Pettigrew and Weisberg (2025) restore sequence-invariance by pooling the prior with each new input multiplicatively before the Jeffrey step,
6/9	The paper borrows the evaluator-with-binary-attributes frame from the statistical-discrimination lineage (Phelps, 1972; Arrow, 1973).	The paper borrows the evaluator-with-binary-attribute frame from the statistical-discrimination lineage (Arrow, 1973; Coate and Loury, 1993).
7/9	instead of stereotypes as representativeness-distortion or selective recall (Bordalo et al., 2016), the sense of stereotype in the current paper is that of a believed cross-attribute association produced by coherent updating on impoverished input, not by a distortion of memory or sampling.	instead of stereotypes as the selective recall of representative types (Bordalo et al., 2016), whose exaggerated associations arise across groups, the stereotype in the current paper is a believed cross-attribute association held by one evaluator and produced by coherent updating on impoverished input, not by a distortion of memory.
8/9	The lineage distinction the concluding section trades on—protection intrinsic to the arithmetic of aggregation, rather than imposed by an enforced constraint on behaviour—goes back at least as far as Becker (1962).	<i>(sentence deleted)</i>
9/9	It is worth contrasting the phenomenon explored in the current paper against the amalgamation paradox (Good and Mittal, 1987), where a real effect present in every subpopulation is erased or reversed by a confound in how the subpopulations are weighted together. Whereas the amalgamation paradox erases a real effect by combining subpopulations under confounded weights, our decision-space result shows the same kind of erasure can occur within a single population – where a first-order share of individuals who are individually affected contributes only a second-order loss in aggregate (Theorem THM:LOS).	It is worth contrasting the phenomenon explored here with the amalgamation paradox (Good and Mittal, 1987), in which a measure of association on a combined table lies outside the range of the same measure on its subtables, because treatment is allocated unevenly across the subpopulations, even when they are equally large. That paradox arises from combining subpopulations. The decision-space result here shows a comparable loss of visibility within a single population, where a first-order share of individuals who are individually affected contributes only a second-order loss in aggregate (Theorem THM:LOS). The audit perspective has a precedent in the economics of discrimination. Heckman (1998) showed that audit-pair estimates of discrimination rest on an assumption about unobserved productivity that “nothing guarantees” (p. 109), so that such an audit “can find discrimination when in fact none exists; it can also disguise discrimination when it is present” (p. 102), and Bohren et al. (2019) identify the source of discrimination by conditioning on the history of evaluations, since in static data, as they note, different sources generate the same patterns of observable behaviour. The obstruction here is of the same kind and arises on the belief side of an audit. The first-order signal in a protected statistic is zero in the population itself, so its silence is a failure of identification and not of estimation, which no sample size repairs, and the identifying variable is the reading sequence, which pooled data discard.

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Why. One entry per sentence group, in manuscript order. BHW/Banerjee: “coarse” belongs to BHW’s binary action, not Banerjee’s continuum; the two share one mechanism, so “on the other hand” goes; the marginals point is the paper’s analogy (M2, M3). Dietrich: Def. 1 is preference aggregation, and with a common prior linear pooling passes the criterion wherever it applies (M4, M5). Epstein/Ortoleva/Cripps (bold sentence): the framing is unsupported for Epstein and Ortoleva; the Cripps claim is false, since under Proposition IMM’s reading the composite satisfies all four axioms, and the footnote goes with it (M6-M9). Pettigrew-Weisberg pool the prior with each input, not successive inputs (M25). Phelps has no binary attribute, so he is dropped from this sentence and can be re-cited in plan 1.C for statistical discrimination in general (M10). BCGS name one mechanism, not two, and “sampling” is not theirs (M12). Becker: the concluding section never draws the distinction, and Becker’s own mechanism is averaging plus a budget constraint, the opposite side of it (“Our statement goes beyond arithmetic”, p. 7; M13); the AFTER deletes the sentence, and an alternative that keeps Becker is in literature/becker1962/README.md. Good-Mittal: their paradox covers amplification and effects created from none, and its cause is uneven allocation, not weighting by population shares (M14). Evidence. BHW.lean, Banerjee.lean, Dietrich.lean, Epstein.lean, Ortoleva.lean (via the 2024 survey), Cripps.lean (<code>order_invariance</code>, <code>composite_AB_eq_bayes</code>), PettigrewWeisberg.lean, Phelps.lean, Arrow.lean, CoateLoury.lean, BCGS.lean, Becker.lean, GoodMittal.lean (<code>piR_amalg_general</code>, <code>equalSize_reversal</code>). The appended paragraph (C.15) carries the identification point of notes/positioning_economics.tex and plan 3.C/3.D into Section 3, with audit P17a (no sample-size claim attributed to Heckman; that clause is the paper’s own consequence of Proposition PRO), H2 (“nothing guarantees” is p. 109), H3 (“can find discrimination” is p. 102) and B37 (BIR credit “different sources” to Fang-Moro, so it is paraphrased) applied. Needs Heckman1998 and Bohren2019 in bibliography.bib (plan B.A).		
C.13 Section 4, the Foster-Greer-Thorbecke sentence		
	This is the standard incidence-versus-intensity pairing of the measurement literature, where a headcount (a count of those past a threshold) and a mean shortfall (average of their distances from it) are first two members of a single parametrised family of indices (Foster et al., 1984).	The two decision statistics pair incidence with intensity, as do the first two poverty measures of Foster et al. (1984), the headcount ratio, the share below a threshold, and the average shortfall below it normalised by the threshold, to which those above it contribute zero.
Why. P_0 is the headcount <i>ratio</i>, a share, and $P_1 = (1/n) \sum g_i / z$ is normalised by the threshold and averaged over the whole population; “average of their distances” is the income-gap measure, which is not in the family (M15). FGT do not use “incidence/intensity”; the AFTER does not attribute it to them. Evidence. FGT.lean (<code>P0_eq_H</code>, <code>P1_eq_popMean_normGap</code>, <code>I_not_decomposable</code>).		
C.14 Appendix, proof of Theorem LOS, the two Tao citations		
1/2	Tonelli’s theorem (Tao, 2011, Corollary 1.7.23) gives	Tonelli’s theorem (Tao, 2011, Theorem 1.7.15), applicable since the integrand is non-negative, gives
2/2	As μ is finite, continuity from above (Tao, 2011, §1.4) yields	As μ is finite and B_c decreases as $c \downarrow 0$, continuity from above (Tao, 2011, Exercise 1.4.23(iii)), applied along any sequence $c_n \downarrow 0$, yields
Why. Corollary 1.7.23 is Fubini-Tonelli; for the nonnegative integrand Tonelli (Theorem 1.7.15, or 1.7.18 for complete measures) is the exact reference. Continuity from above is Exercise 1.4.23(iii), stated for sequences; the proof takes c to 0 over a continuum (M16). Numbering is from the author’s preprint of Tao; confirm against the printed book. Evidence. Tao.lean (<code>thm_1_7_15</code>, <code>ex_1_4_23_iii</code>, <code>measure_band_tendsto_zero</code>, <code>los_step4</code>).		

Section E: the archived plan’s entries, re-issued and corrected

The entries of notes/manuscript_change_plan_asof_2026-09-30.md that the author’s draft did not already apply, each corrected against the audit and passed through the discipline gate. Order of application: E.10 (Propositions ORD and ADJ) before anything that cites them; E.14 (bibliography) before E.1, E.5, E.7 and C.15; E.11 before E.12.

Part	BEFORE (current manuscript, or the anchor for an insert)	AFTER (proposed)
E.1	Introduction, new paragraph after the hiring-panel paragraph (was 1.C)	

Part	BEFORE (current manuscript, or the anchor for an insert)	AFTER (proposed)
	<i>New paragraph(s) after the paragraph containing “with the difference set by arrival sequence.”</i>	Which impression prevails matters beyond the single judgement. The impression-formation literature disagrees on the direction. Asch (1946) reports that the earlier terms set the direction in which later ones are read, and the review of Hogarth and Einhorn (1992) finds primacy, recency or no effect depending on the characteristics of the task. Under either direction the sequence in which cues arrive is not neutral, and it need not be assigned by chance. A candidate who comes through a referral is met first through the letter and then through the credential, while a candidate who applies unsolicited is met first through the credential. If referral is more common in one group than in another, the two groups differ in the share λ of evaluators who read the credential first. Two groups presenting the same evidence to evaluators with the same priors and the same preferences then receive different mean beliefs, and the difference is the sequence effect multiplied by the difference in their shares. A gap of that kind is not statistical discrimination in the sense of Phelps (1972) and Arrow (1973), which rests on different prior beliefs about the groups, and it is none of the three sources that Bohren et al. (2019) distinguish, correct beliefs, biased beliefs and preferences, since none of these differs between the groups here. Which group it favours depends on whether the later or the earlier impression prevails, which Proposition PROP:ADJ reads from the marginals. The gap also inherits the classification of this paper. It appears in the marginal probabilities and in the share of decisions changed, and an audit of the believed association does not register it.
Why. Plan 1.C with audit P4 (“prior beliefs about the groups”), D13 (BIR’s three sources named rather than “none of the sources”, since their impartial type discriminates with correct beliefs and no animus), A24 (Asch’s “early terms dominate” softened to his own “direction” account) and H6 (the primacy/recency/no-effect finding is HE’s review of other studies) applied. Presupposes E.10 (Proposition ADJ) and the Bohren2019 bib entry. Goes after C.5’s sentence. Evidence. The identity $\text{mean}(\lambda) - \text{mean}(\lambda') = (\lambda - \lambda')(PJ_AB - PJ_BA)$ is exact, verify_ORD.py step 9; Asch.lean; HogarthEinhorn.lean; BohrenImas-Rosenberg.lean.		
E.2 Setup 2.1, what an impression is (was 2.A)		
	yields an impression (a target marginal) $q = (q_0, q_1)$ on A ’s partition and a cue on B yields an impression $r = (r_0, r_1)$ on B ’s partition—with no cue bearing on a joint event.	yields an impression (a target marginal) $q = (q_0, q_1)$ on A ’s partition, with q_i the credence that $A = i$ and $q_0 + q_1 = 1$, and a cue on B yields an impression $r = (r_0, r_1)$ on B ’s partition, with r_j the credence that $B = j$ and $r_0 + r_1 = 1$, and no cue bears on a joint event. An impression is thus a probability distribution on one attribute, on the same scale as the prior marginal it replaces, $(\alpha, 1 - \alpha)$ for q and $(\beta, 1 - \beta)$ for r .
Why. Plan 2.A as written; the manuscript never says that q_0 belongs to $A = 0$, that $q_0 + q_1 = 1$, or that an impression is on the prior marginal’s scale. The dash construction is replaced by a clause. Evidence. Notation only; every sympy script and Lean file uses $q_1 = 1 - q_0$, $r_1 = 1 - r_0$.		
E.3 Setup 2.2, the worked example, and why a soft cue differs from a hard one (was 2.B, extended)		

Part	BEFORE (current manuscript, or the anchor for an insert)	AFTER (proposed)
	<i>New paragraph(s) after the paragraph containing “The gap is the difference between either sequence-posterior and the benchmark,”</i>	A worked example. Let $\alpha = \beta = \frac{1}{2}$ and $c = \frac{1}{20}$, so that the prior table is $P = \begin{pmatrix} .30 & .20 \\ .20 & .30 \end{pmatrix}$, each trait judged as likely absent as present and the two believed to go together. The credential delivers $q_0 = \frac{1}{5}$ and the letter $r_0 = \frac{7}{10}$. After the credential alone the table is $\begin{pmatrix} .12 & .08 \\ .32 & .48 \end{pmatrix}$. The A-marginal is $(.20, .80)$, as delivered, and the B-marginal has moved from $(.50, .50)$ to $(.44, .56)$ although no cue about B has been read. Reading the letter next resets the B-marginal to $(.70, .30)$ and moves the A-marginal from $.20$ to $P_{AB}^J(A=0) = .234$. In the reverse sequence the A-marginal ends at $.20$, as delivered, and the B-marginal at $P_{BA}^J(B=0) = .643$ in place of $.70$. The benchmark gives $P^B(A=0) = .227$ and $P^B(B=0) = .647$. The sequence effect is therefore $.034$ on the A-marginal and $.057$ on the B-marginal. On the believed association it is $.0002$, since $\text{assoc}(P_{AB}^J) = .0273$ and $\text{assoc}(P_{BA}^J) = .0271$, smaller than either marginal effect by a factor above 150. The odds ratio equals $9/4$ for the prior, for both sequences and for the benchmark. Against the benchmark, sequence AB departs by $.053$ on the marginal read last, by $.007$ on the marginal read first, and by $.002$ on the association. Why a soft cue differs from a hard one. Suppose instead that the letter were a hard cue, a proposition F the panel becomes certain of, with a likelihood fixed before any belief is formed, say $P(F B=0)/P(F B=1) = 7/3$, and the credential likewise a proposition E with ratio $1/4$. Conditioning on E and then on F is conditioning on $E \cap F$, so the two sequences give the same belief, which is the benchmark P^B, with A-marginal $.227$ in either sequence. The soft letter delivers the credence $r_0 = \frac{7}{10}$ instead. The factor that credence implies is the ratio of the odds it delivers to the odds it meets, $7/3$ when the letter is read first against the prior marginal $\frac{1}{2}$, but $98/33$ when it is read after the credential, against the marginal $.44$ that the credential has already moved. The panel knows the credence the letter left, not what that credence would have been at another prior, and so cannot extract a portable factor from it. That dependence of the implied factor on the belief it meets is the whole of the sequence effect, and it vanishes at $c = 0$, where the credential leaves the B-marginal where it was and the two implied factors coincide.
Why. Plan 2.B, with the colon removed, plus the passage the author asked for on 2026-09-30, saying exactly what is special about a soft cue. A hard cue is a proposition with a belief-independent likelihood, so conditioning is defined and commutes; a soft cue fixes a credence, and the factor that credence implies is matched to the marginal in force when it arrives ($7/3$ read first, $98/33$ read second). On one cue the two readings agree (Proposition IMM). This replaces the counterfactual-likelihood argument the audit found contradicted by Jeffrey’s definition of a Bayes factor (J2). With $c = 1/20$ against a bound of $1/4$ the example illustrates the ordering of magnitudes, not the asymptotic rates. Evidence. Every number in both paragraphs is asserted in <code>sympy/verify_example.py</code> (18/18) and <code>sympy/verify_soft_vs_hard.py</code> (11/11, added 2026-09-30, registered in <code>run_all.py</code>); the mechanism is <code>Cripps.lean composite_AB_eq_bayes</code> and <code>DiaconisZabell.lean thm21</code>.		
E.4 Setup 2.3, Assumption 2 (was 2.C)		
	The cues are bundled, correlated, and soft, each delivering a credence on an attribute’s partition, not a decisive cue.	The cues are bundled, correlated, and soft, each delivering a credence on an attribute’s partition, not a decisive cue, with $0 < q_0 < 1$ and $0 < r_0 < 1$.
Why. Plan 2.C, with the colon replaced by a clause. Evidence. Notation only.		
E.5 Related literature, new opening paragraphs on the commutativity literature (was 3.A)		

Part	BEFORE (current manuscript, or the anchor for an insert)	AFTER (proposed)
	While sequence-dependence has been of interest to empirical psychology at least since Asch (1946), the problem seems to have received interest in economics primarily through the observational learning literature.	The closest literature concerns the commutativity of probability kinematics itself. Diaconis and Zabell (1982) give the condition under which two successive Jeffrey revisions lead to the same belief, and Hawthorne (2004) names the full-adoption premise, which the present paper adopts, the Amnestic Update-Factor Thesis, crediting that criterion to them. His survey also fixes the vocabulary. Extensions of Jeffrey updating to sequences differ in what an experience is taken to deliver, new probabilities for its own basis directly, a multiplicative factor applied to the belief it meets, or a ratio between basis sentences that the prior does not constrain. The model used here takes the impression to deliver a credence, with the most recent one fixing its attribute outright, and the benchmark P^B of Section SEC:JEFFREY has the form of his extended update formula built on multiplicative factors. The comparison drawn throughout is therefore between his amnestic model and his factor models, and the identification questions this paper raises are the ones the amnestic model brings with it and the factor models do not, since factor revisions on distinct attributes commute. What Section SEC:JEFFREY calls a Bayes factor is, up to normalisation, his normed-likelihood factor, and as a ratio of two such factors it is his likelihood-ratio factor, which he too calls a Bayes factor; on a two-element basis all induce the same revision. The premise is disputed on two different grounds. Döring (1999) objects on normative grounds, that a sequence dependence of this size is unjustified in an account of rational belief change. Hawthorne (2004, p. 115) objects on psychological grounds and declines to settle them, asking “Which extension of Basic Jeffrey Updating is the more plausible model of human agents? I’m a logician, not a psychologist.” He judges amnestic updating less plausible than the factor-based extensions, “because it seems unlikely that we dismiss previous experiences so completely”, and turns to the normative properties of the competing models, where his interest lies. The ground on which the paper adopts the premise is information-theoretic, since Jeffrey’s rule is the revision that moves the prior least in I-divergence while taking on the delivered credence (Diaconis and Zabell, 1982). An objection of Hawthorne’s form is a reason to measure the degree of adoption, not a reason to discard the premise. With two cues that degree has exactly two ends, the impression read last prevailing and the impression read first prevailing, and Proposition PROP:ADJ recovers where between those ends an evaluator sits, from the rating of one trait read before and after the second cue and against what that cue delivers. Two responses to the resulting sequence dependence have been made. One treats it as a defect of the framework. Döring (1999) argues that the sequence effect can be pronounced enough to call for an adjustment that successive Jeffrey steps cannot supply, though a single Jeffrey step from the original prior can. The other re-describes the input so that the effect disappears. Field (1978) reparametrises the update so that its input is a portable factor rather than a delivered credence, and Wagner (2002) shows that when identical learning is represented by identical Bayes factors, sequential revisions commute; Garber (1980) objects that a portable factor compounds implausibly under repetition. The present paper takes neither route. It accepts the sequence dependence that the delivered-credence reading entails, and asks how far the two sequences disagree in each statistic of the resulting belief. That the disagreement is uneven across statistics, first order in the marginals and the share of decisions changed and second order in the believed association and the surplus-weighted loss, is what makes the debate answerable by measurement rather than by introspection about how completely an evaluator dismisses an earlier impression. The insensitivity of the association is moreover not peculiar to the extension adopted here. Each of the three extensions revises a basis by multiplying its cells by a factor that

Part	BEFORE (current manuscript, or the anchor for an insert)	AFTER (proposed)
	Why. Plan 3.A with the audit applied. D1/P3, Doring’s objection is normative and Hawthorne’s psychological, so the two are separated and “disputed on psychological grounds” goes. P2, Doring’s own remedy is a single Jeffrey update from the original prior. M21, “the two ends of his taxonomy” becomes “his amnesic model and his factor models” (NL is the middle model), and the Bayes-factor identification names the normalisation and his own use of “Bayes factor” for LR factors (his note 20). “Order” becomes “sequence” throughout (the manuscript’s convention). Two colons and one dash removed. The Hawthorne quotations are verbatim from pp. 115-116; the second drops “not because it is un-Bayesian, but” before “because”, which leaves his stated reason unchanged. Needs Doring1999 and Garber1980 in bibliography.bib (E.14). Evidence. Doring.lean (fig2, gap_tends_to_one); Hawthorne.lean (amnesic_thesis, factorUpdate_comm, NL_jeffrey, med_NL_denominator); Field.lean (field_eq7_comm); Wagner2002.lean (thm31); Garber.lean (garber_table); DiaconisZabell.lean (thm51_KL_eq_iff); LemmaSEP.lean.	
E.6	Related literature, Asch as evidence (was 3.B)	
	While Bayes-factor updating predicts no sequence effect anywhere (Wagner, 2002), the belief-adjustment accounts of order effects (Hogarth and Einhorn, 1992) focus on a single evaluative anchor. What the paper highlights is that such experimental analyses must also factor in the cross-attribute association – thus requiring the elicitation of a full posterior table rather than a single rating produced by step-by-step or end-of-sequence response modes.	While Bayes-factor updating predicts no sequence effect anywhere (Wagner, 2002), the experimental tradition has measured the phenomenon entirely in marginals. Hogarth and Einhorn (1992) track a single evaluative anchor. Asch (1946) is more generous and more instructive. His Experiment VI reports, for each of eighteen traits, the proportion of subjects choosing it over its opposite for the person described, separately for the two reading sequences. That the sequence effect differs in size across statistics is already visible there, although the differences Asch reports are all among marginals, which the paper treats alike. Between the sequences, <i>restrained</i> moves from 64 to 9 per cent and <i>good-looking</i> from 74 to 35, while <i>serious</i> moves from 97 to 100, <i>persistent</i> from 82 to 87 and <i>reliable</i> from 84 to 91. Some statistics swing enormously and others scarcely move, and Asch attributes the difference to the content of the terms rather than to their position. Yet eighteen marginals separate full adoption from partial adoption no better than one. Separating them needs the rating of one trait before and after the second cue and against what that cue delivers (Proposition PROP:ADJ), and the check-list asks whether a trait fits and never how strongly two traits are believed to go together beforehand, so the prior association cannot be formed from such data at all. What the paper adds is therefore not that more ratings are needed but that a second object must be elicited. Seeing the sequence effect requires only a marginal read in both sequences, which Asch has; locating the mechanism behind it requires the prior association as well, which he does not.
	Why. Plan 3.B with the audit applied. A11, the check-list is a forced choice between opposite pairs. A15, Asch does give an account of the uneven effects, by content. P13, what is never elicited is the prior association, not the joint. The ADJ clause now states what the recovery formula uses, three ratings of one marginal, not a marginal across values of c. “amnesic updating” becomes “full adoption”. Colon removed. All eight percentages are correct and in the right columns. Evidence. Asch.lean (quoted_all, d_neg_exactly, stimulus_disjoint_checklist); Anchoring.lean (dampedB_deviation, the_identification).	
E.7	Related literature, Bohren-Imas-Rosenberg, after the identification paragraph (was 3.D)	

Part	BEFORE (current manuscript, or the anchor for an insert)	AFTER (proposed)
	<i>New paragraph(s) after the paragraph containing “contributes only a second-order loss in aggregate (Theorem THM:LOS).”</i>	A current formulation of the same problem is given by Bohren et al. (2019), who distinguish discrimination arising from correct beliefs, from biased beliefs, and from preferences, and identify the source from how discrimination evolves along a history of evaluations. Their map from partiality to behaviour is Bayesian. A belief gap between groups is transmitted to evaluations, attenuated by the precision of the signal and along the history, and it vanishes as judgement becomes perfectly objective. Under the reading adopted here that map does not hold. A Jeffrey step sets the marginal of the attribute it addresses to the delivered credence, so an evaluator who takes the impression of quality last evaluates two workers alike whatever her prior beliefs about their groups, and the belief gap is silenced without any gain in objectivity. Under partial adoption with weight ω on the impression, a fraction $1 - \omega$ of the belief gap survives (Proposition PROP:ADJ). What credence-input updating does is therefore not to add a further source of discrimination to their three, but to relocate where partiality must sit in order to act. Lodged in the prior it is silenced, lodged in the impression it passes through untouched, and their framework has no parameter for the latter.
Why. Plan 3.D with the audit applied. P17b, the belief gap is also attenuated along the history (their Proposition 2), so “vanishes only as” becomes “vanishes as”. Colon removed. Goes directly after C.15’s paragraph, which carries the Heckman precedent that plan 3.C proposed (3.C is superseded by C.15). Presupposes E.10 and the Bohren2019 entry. Evidence. <code>check_pinning_kills_partiality.py</code> (7/7, $D = 0$ exactly for arbitrary group priors and covariances); <code>Anchoring.lean</code> <code>dampedB_deviation</code>; <code>BohrenImasRosenberg.lean</code> (<code>prop1_decreasing</code>, <code>prop2_decreasing</code>, <code>endo_gap</code>).		
E.8 Section 4, title and opening (was 4.A)		
1/2	<i>Section title: Sequence effects at the individual level</i>	<i>Section title: Which statistics register the sequence</i>
2/2	Since Bayesian conditioning is sequence-independent by definition, we measure sequence effects with respect the Bayesian benchmark using two primitive statistics – gap and score. As discussed in Section SEC:JEFFREY, the <i>gap</i> is a distance from a Bayes-factor benchmark P^B for the evaluator (see Definition DEF:INDGAP). The second primitive, <i>score-gap</i> $s(P) = \langle v, P \rangle$ defined against the sequence-free reference P^B alongside with a threshold τ . As we show in Section SEC:INDIVIDUAL_PRIMITIVES, both <i>gap</i> and <i>score-gap</i> are first-order to the evaluator. The belief-statistics and decision-statistics follow from the two primitive statistics. The two belief-statistics that we are interested in are the <i>marginal probability</i> and <i>believed cross-attribute association</i> . The decision-statistics that we discuss for the individual evaluator and the population aggregate are <i>surplus-weighted loss</i> $L(c)$ and a <i>decision-flip</i> . The decision-flip is counted in the population to determine the <i>share</i> of those evaluators whose decision is changed by the sequence. The loss is then simply an aggregation of $ u $ over this share. How these quantities are carried over in aggregate are detailed in Section SEC:AGGREGATION.	This section asks, of each statistic of a single evaluator’s belief, whether the two reading sequences move it at first order in the prior covariance c or only at second order. Two primitive statistics carry the comparison with the sequence-free benchmark P^B of Section SEC:JEFFREY, the <i>gap</i>, the distance of the evaluator’s belief from P^B (Definition DEF:INDGAP), and the <i>score-gap</i> $s(P) = \langle v, P \rangle$, the same distance read through the decision weights and set against a threshold τ. Section SEC:INDIVIDUAL_PRIMITIVES shows that both are first order. The belief statistics and the decision statistics are read from the same belief. The belief statistics are the <i>marginal probability</i> of each attribute and the <i>believed cross-attribute association</i>; the decision statistics are the surplus-weighted loss $L(c)$ and the decision <i>flip</i>, the indicator that the sequence changes the evaluator’s decision. Whether a statistic registers the sequence is settled here, for one evaluator; Section SEC:AGGREGATION shows that averaging over evaluators who met the cues in either sequence leaves the classification unchanged across an open set of priors and every interior mixture.
Why. Plan 4.A, colon removed. The section is organised on the statistic rather than on the individual-versus-aggregate axis, which ORD and PRO show to be inert. The third paragraph (why these statistics; FGT, corrected by C.13) is unchanged. Evidence. DIV and SCR (both primitives first order); ORD and PRO (the classification is settled before averaging).		
E.9 Section 5, title and opening (was 5.A)		
1/2	<i>Section title: Aggregated Effects of Cue-Sequence</i>	<i>Section title: Averaging over sequences does not change the classification</i>

Part	BEFORE (current manuscript, or the anchor for an insert)	AFTER (proposed)
2/2	We now describe how the effect of arrival order on statistics discussed in the previous section fares in the population aggregate. Throughout this section we consider a population mixing the two encounter sequences: a fraction $\lambda \in [0, 1]$ of evaluators meet the A -first (the credential cue in the example) and the rest B -first (the letter), so the mean belief under Jeffrey conditioning is $\bar{P}_\lambda = \lambda P_{AB}^J + (1 - \lambda) P_{BA}^J$. As it turns out, what makes a statistic second order or first order is largely the nature of the statistic itself rather than the mechanism of aggregation.	The previous section classified statistics for one evaluator. This section shows that the classification survives averaging over a population in which a fraction $\lambda \in [0, 1]$ of evaluators meet the A -cue first (the credential in the example) and the rest the B -cue first (the letter), so that the mean belief under Jeffrey conditioning is $\bar{P}_\lambda = \lambda P_{AB}^J + (1 - \lambda) P_{BA}^J$. Across an open set of priors, what makes a statistic first order or second order is the statistic itself and not the mixture λ , since the mean belief lies in the plane through P^B spanned by the leading directions of the two sequences, and a statistic's order is fixed by its differential on that plane (Propositions PROP:ORD and PROP:PRO).
Why. Plan 5.A, colon removed. “Largely the nature of the statistic itself” hedged an exact result. Presupposes E.10.		
Evidence. PropPRO.lean (annihilator_eq_span, propPRO_uniqueness); PropORD.lean.		
E.10 Section 5, Propositions ORD and ADJ after the definition of protection (was 5.B)		

Part	BEFORE (current manuscript, or the anchor for an insert)	AFTER (proposed)
1/2	New paragraph(s) after the paragraph containing “unprotected if that distortion is first order, $\mathcal{O}(c)$.”	The distortion compares the population average with the benchmark. The same classification can be reached without either ingredient, by reading the sequence effect of Definition DEF:SEQEFFECT through the statistic itself. For a smooth statistic F the sequence effect read through it is $F(P_{AB}^J) - F(P_{BA}^J)$, and it involves neither the benchmark P^B nor the mixing weight λ. Proposition ORD (between-sequence contrast). <i>For any weight v,</i> $\langle v, P_{AB}^J - P_{BA}^J \rangle = c(\kappa \langle v, R_1 \rangle - \kappa' \langle v, R_2 \rangle) + \mathcal{O}(c^2),$ <i>with $\kappa, \kappa', R_1, R_2$ as in Proposition PROP:DIV. The sequence effect on the A-marginal is $c\kappa' + \mathcal{O}(c^2)$ and on the B-marginal $-c\kappa + \mathcal{O}(c^2)$; each is $\Theta(c)$ generically. The sequence effect on the believed association is $\mathcal{O}(c^2)$ exactly. Across an open set of priors, a smooth statistic has a second-order sequence effect if and only if it is protected (Definition DEF:PROTECTION); both conditions hold exactly when the differential of the statistic at independence annihilates $\text{span}\{R_1, R_2\}$ (Proposition PROP:PRO below). Nothing in the display involves P^B or λ. The classification is decided by the two reading sequences, and every mixture $\bar{P}_\lambda$ inherits it.</i> <i>Proof.</i> By Proposition PROP:DIV(ii), $P_{AB}^J - P_{BA}^J = c(\kappa R_1 - \kappa' R_2) + \mathcal{O}(c^2)$; applying $\langle v, \cdot \rangle$ gives the display. With $R_1 = q \otimes (1, -1)$ and $R_2 = (1, -1) \otimes r$, $\langle \mathbf{1}_{A=1}, R_1 \rangle = 0, \quad \langle \mathbf{1}_{A=1}, R_2 \rangle = -1, \quad \langle \mathbf{1}_{B=1}, R_1 \rangle = -1, \quad \langle \mathbf{1}_{B=1}, R_2 \rangle = 0,$ giving $c\kappa'$ and $-c\kappa$. For the association, $\langle \nabla \text{assoc}(q \otimes r), R_1 \rangle = \langle \nabla \text{assoc}(q \otimes r), R_2 \rangle = 0$ (proof of Proposition PROP:PRO), so its sequence effect is $\mathcal{O}(c^2)$. For the equivalence, with $M_\lambda = \lambda\kappa R_1 + (1 - \lambda)\kappa' R_2$ from Proposition PROP:DIV(i) and $\nabla F := \nabla F(q \otimes r)$, $\langle \nabla F, \Delta_{\text{seq}} \rangle = \kappa \langle \nabla F, R_1 \rangle - \kappa' \langle \nabla F, R_2 \rangle, \quad \langle \nabla F, M_\lambda \rangle = \lambda\kappa \langle \nabla F, R_1 \rangle + (1 - \lambda)\kappa' \langle \nabla F, R_2 \rangle;$ across an open set of priors κ and κ' vary independently, so either vanishes identically exactly when $\langle \nabla F, R_1 \rangle = \langle \nabla F, R_2 \rangle = 0$. $\square$ The two sequences so far adopt each delivered credence in full. Suppose instead that the second cue is adopted only in part, so that after the first Jeffrey step the response to the second cue is a Jeffrey step on its partition to the target $(1 - \omega)m + \omega r_1$, where m is the marginal the first step left and $0 \leq \omega \leq 1$ is the adoption weight of Section SEC:SETUP. The rule is the averaging form of the belief-adjustment model, $S_k = (1 - w_k)S_{k-1} + w_k s(x_k)$ (Hogarth and Einhorn, 1992, Eq. 4), applied to the second cue only, which is their end-of-sequence form for two items with a constant weight, and embedded in the joint law by a Jeffrey step. Write P_{AB}^ω and P_{BA}^ω for the two sequences under this rule; $\omega = 1$ recovers P_{AB}^J and P_{BA}^J, and $\omega = 0$ ignores the second cue. Let P^A and P^B denote the belief after the A-cue alone and after the B-cue alone. Proposition ADJ (adoption weight). <i>For every c and every ω, in sequence AB,</i> $P_{AB}^\omega(B=1) - r_1 = (1 - \omega)[P^A(B=1) - r_1],$ $P_{AB}^\omega(A=1) - q_1 = \omega[P_{AB}^J(A=1) - q_1],$ $P_{AB}^\omega(A=1) - P_{BA}^\omega(A=1) = \omega[P_{AB}^J(A=1) - q_1] + (1 - \omega)[q_1 - P^B(A=1)].$ <i>Hence the last read marginal equals its delivered credence identically in c exactly when</i>

Part	BEFORE (current manuscript, or the anchor for an insert)	AFTER (proposed)
2/2	A particularly surprising finding is that for a smooth statistic of the belief, the answer is settled before any averaging over the aggregate takes place.	A particularly surprising finding is that for a smooth statistic of the belief, the answer is settled before any averaging over the aggregate takes place (Proposition PROP:ORD).
Why. Plan 5.B, mathematics unchanged, prose corrected. P10/P11, the one-sided damping is Hogarth-Einhorn’s own end-of-sequence form (their Eq. 8) for two items with a constant weight, not a construction of ours; what is new is the two-attribute Jeffrey embedding. The adoption weight is now defined in Setup (author’s draft; C.9), so the lead-in refers there. Colons before displays are kept; colons in prose are replaced. The closing example sentence is plan 2.B’s optional sentence. Both propositions must go in before E.1, E.5, E.6, E.7, E.9, E.11, E.12 and C.5, which cite them. Evidence. PropORD.lean (lemmaORD_gap, propORD_Amarg, propORD_Bmarg, propORD_const); verify_ORD.py (25/25); Anchoring.lean (dampedB_deviation, routeDamped_mA1_deviation, orderEffect_damped_mA1, orderEffect_damped_at_indep, dampedB_at_one, dampedB_at_zero, routeDamped_at_zero_pins_A); check_zero_slope_identification.py (25/25); verify_example.py (the omega = 1/2 read-back); HogarthEinhorn.lean (oneSided_eq_eq8).		
E.11 Scope, rival mechanisms (was 6.A)		
	<i>New paragraph(s) after the paragraph containing “is outside the scope of stated results.”</i>	A separate scope question concerns rival mechanisms rather than broken assumptions. Sequence effects have an explanation that competes with full adoption. In the belief-adjustment model of Hogarth and Einhorn (1992), when the first cue is adopted in full and later cues only in part the first impression is protected, and at zero adoption of later cues it is never moved, whereas when every cue is adopted only in part from a prior anchor the same model predicts recency. That first account and full adoption point in opposite directions, since the protected impression is the first one there and the last one here. An observable separates them. Under partial adoption of the second cue with weight ω (Proposition PROP:ADJ), the marginal that ignores the prior association is the last-read one at $\omega = 1$, the first-read one at $\omega = 0$, and neither in between. A reading on which neither marginal ignores the association is consistent with an interior weight and with the benchmark alike; the two are separated by the sequence comparison, under which the benchmark shows no sequence effect at any c and an interior weight shows one already at $c = 0$ (Proposition PROP:ADJ). Since every rule here is a separable reweighting, all of them carry the prior’s odds ratio (Lemma LEM:SEP), so the odds ratio separates none of them and the two marginals are the only statistics that do. Three cautions bound the claim. The classification covers this one-parameter family, not every conceivable mechanism. Asch (1946) is evidence that sequence moves marginals, six stimulus terms read in two sequences moving the proportions on eighteen response traits, and not evidence for either endpoint, since his own account is that early terms set a direction for the reading of later ones, so that what a later cue delivers depends on its position, a mechanism outside the family here, in which each cue delivers the same credence in either position. And full adoption is a substantive commitment rather than a consequence of the level reading. Hawthorne (2004) objects that it “seems implausible that the most recent experience or non-propositional state should completely dictate belief strengths for basis sentences, with no regard for the import of previous experiences or states”. The objection applies here, and his own two-basis example shows the overwriting explicitly, so that the credential’s implication for trustworthiness is erased once the letter fixes that attribute. The adoption weight is his objection stated as a parameter, and the reply offered here is not that the objection misses but that the weight is recovered from three ratings of one marginal rather than assumed.

Part	BEFORE (current manuscript, or the anchor for an insert)	AFTER (proposed)
	Why. Plan 6.A with the audit applied. P10, the belief-adjustment model protects the first impression only when the first cue is the anchor and later cues are partly adopted; damping every cue gives recency, so the rival is stated as that case. P14, Experiment VI has six stimulus terms and eighteen response traits. P9, Hawthorne's objection follows his two-basis medical example, so the claim that his illustration misses a one-cue-per-attribute model is dropped. The recovery uses three ratings of one marginal (plan's own correction). Colon removed. Goes after the scope paragraph, before E.12. Evidence. HogarthEinhorn.lean (eq8_estimation_first_dominates, appB_recency, twoAttr_oneSided_orderEffect); Anchoring.lean; Asch.lean (seriesA_length, checkListI_length); Hawthorne.lean (med_overwrite; quotation pp. 98-99 verbatim).	
E.12	Scope, the two channels and the rubric prediction, with the settings table (was 6.B, from interior_omega.tex)	

Part	BEFORE (current manuscript, or the anchor for an insert)	AFTER (proposed)						
	<i>New paragraph(s) after the paragraph containing “is outside the scope of stated results.”</i>	The two channels. Sequence dependence has two channels in this setting, and the results of Sections SEC:INDIVIDUAL and SEC:AGGREGATION concern one of them. One is the association channel. A cue on one attribute moves the belief about the other through the prior association, so the second cue meets a belief the first has already changed. The association channel’s effect is first order in c and vanishes at independence, where each cue sets its own marginal and touches nothing else (Proposition PROP:IMM). The other is the position channel. If the cue read second is adopted only in part, then which cue is weakened depends on the sequence, and the two sequences differ even at $c = 0$, by $(1 - \omega)(\alpha - q_0)$ on the competence marginal and $-(1 - \omega)(\beta - r_0)$ on the trustworthiness marginal. The position channel is not a feature of updating on delivered credences. A Bayes-factor update that gives the second cue’s factor the weight ω is sequence-dependent in the same way, since commutation requires the same factor in either position (Wagner, 2002). Under full adoption the position channel is absent and the association channel is the only one, which is the setting of the results. Under partial adoption the position channel is present at $c = 0$ and dominates the association channel near it, so the classification by order in c no longer measures what it was built to measure; the marginals differ at zeroth order, the believed association still does not differ between sequences at $c = 0$, and the mean belief of a population mixing the two sequences acquires a cross-product association of $-\lambda(1 - \lambda)(1 - \omega)^2(\alpha - q_0)(\beta - r_0)$ that no member holds. What the paper says about the position channel is confined to Proposition PROP:ADJ, which recovers its strength from the ratings. The position channel is a temporal bias of the observer, present whether or not the attributes are believed related, while the association channel runs through what one cue implies about the other and exists only when they are. Table TAB:SETTINGS sets the two settings side by side. Opposite predictions about one procedure. Whether a procedure removes the sequence effect depends on which channel it removes. Consider rubric scoring, in which the rating on each attribute is fixed by the document on that attribute and a set scale, so that what each document delivers no longer depends on what was read first. Under full adoption the rubric removes the position channel and leaves the association channel untouched. The recorded scores carry the delivered credences in either sequence, the belief about the attribute read second agrees with its score, and the belief about the attribute read first sits at cK from its score (Proposition PROP:DRF), so the sequence effect lives in that belief and in the decision made from it, vanishes when the attributes are believed unrelated (Proposition PROP:IMM), and is invisible in the scores and in the association between them. Under partial adoption the rubric removes the position channel from the record only. The belief about the attribute read second sits at $(1 - \omega)[P^A(B=1) - r_1]$ from its score (Proposition PROP:ADJ), which at independence equals $(1 - \omega)(r_0 - \beta)$, so the two reading sequences differ even when the attributes are believed unrelated. The two settings therefore predict opposite things about the same procedure, and the difference is observable. In the rubric setting r_1 is on the record, so Proposition PROP:ADJ needs only the belief about trustworthiness after the credential alone and after both documents to say which prediction holds. <table> <tr> <th></th><th>Full adoption, $\omega = 1$ (association channel only)</th><th>Partial or no adoption, $\omega < 1$ (position channel operates)</th></tr> <tr> <td>Where the sequence dependence arises</td><td>In the rule. The inputs q, r are the same in either position; the sequence matters because two rigid steps on correlated parti-</td><td>In the inputs. What the second cue delivers depends on its position; the rule adds nothing new at $c = 0$.</td></tr> </table>		Full adoption, $\omega = 1$ (association channel only)	Partial or no adoption, $\omega < 1$ (position channel operates)	Where the sequence dependence arises	In the rule. The inputs q, r are the same in either position; the sequence matters because two rigid steps on correlated parti-	In the inputs. What the second cue delivers depends on its position; the rule adds nothing new at $c = 0$.
	Full adoption, $\omega = 1$ (association channel only)	Partial or no adoption, $\omega < 1$ (position channel operates)						
Where the sequence dependence arises	In the rule. The inputs q, r are the same in either position; the sequence matters because two rigid steps on correlated parti-	In the inputs. What the second cue delivers depends on its position; the rule adds nothing new at $c = 0$.						

Part	BEFORE (current manuscript, or the anchor for an insert)	AFTER (proposed)
Why. Plan 6.B, whose text lived in notes/interior_omega.tex sections 6 and 2 (approved 2026-09-27), carried into the manuscript with the audit applied. “Order” becomes “sequence” for reading order throughout, including the table’s first row; Weisberg’s name is kept out of the table (his rule-versus-inputs framing is not his, audit D10); “The first is / The second is” becomes “One is / The other is” to avoid colliding with the first and second cue; “This effect” is given its noun; proposition names become references; the table is a booktabs table (the manuscript loads booktabs and array, not longtable). Goes after E.11. The decision row of interior_omega’s independence table (share and loss at interior omega) is inferred, not verified, and is left out, as decided 2026-09-27. Every remaining quantity is a verified row. Evidence. sympy/verify_interior_omega.py (23/23: rows 1-6 of the independence table, the (1-omega) and (1-omega)^2 factors, the position-scaled Bayes-factor update at row 5); Anchoring.lean (orderEffect_damped_at_indep); LemmaSEP.lean; PropDRF; the rubric prediction is Propositions DRF, IMM and ADJ read together.		
E.13 Back matter, AI declaration (was B.B)		
	Proposition PROP:PRO (uniqueness of the protected statistic), which was then independently re-derived	Propositions PROP:PRO (uniqueness of the protected statistic), PROP:ORD (between-sequence contrast) and PROP:ADJ (adoption weight), which were then independently re-derived
Why. Plan B.B. Propositions ORD and ADJ have the same provenance as PRO, so the declaration names them once E.10 is applied. Evidence. None needed.		
E.14 Bibliography entries the Section E texts need (was B.A)		
	No manuscript text; see Why.	
Why. Needed by E.1, E.5, E.7 (Bohren2019, Doring1999, Garber1980) and C.15 (Heckman1998, Bohren2019). Hawthorne2004 is already in bibliography.bib. All six proposed entries are in notes/manuscript_corrections_extra.bib and move to bibliography.bib on approval. Zhao-Osherson is cited only in the introduction (author’s draft) and is already in the bibliography. Evidence. Bibliographic details verified against the Drive copies where printed (Doring: Phil. Sci. 66 (Proceedings) S379-S389; Garber: 47(1) 142-145; Heckman: JEP 12(2) 101-116); Bohren et al.’s AER pages are from the reference lists of later papers, the Drive copy being the working paper.		

C.12 and E.14: bibliography entries the AFTER texts need

Hawthorne2004 and ZhaoOsherson2010 are already in bibliography.bib. Six more are needed by C.4’s optional sentence, C.6, C.10, C.15, E.1, E.5 and E.7 (Jeffrey 2004, Benjamin et al. 2019, Heckman 1998, Bohren et al. 2019, Döring 1999, Garber 1980); they are in notes/manuscript_corrections_extra.bib so that this document renders them, and move to bibliography.bib on approval. The Drive copy of Jeffrey (2004) is the November 2002 draft, so C.6 and C.10 cite the chapter only.

Section D: where each entry of the archived plan (notes/manuscript_change_plan_asof_2026-09-30.md) now lives

Entry	Subject	Status and what it needs
0.A	Abstract	Superseded by the author’s rewrite and C.1.
1.A, 1.B, 1.D, 1.E	Introduction	Superseded by the author’s rewrite; what survives is corrected in C.2-C.8. The ORD sentence of 1.E is still unapplied and presupposes E.10.
1.A2	Two-channel prelude	Applied by the author; corrected by C.4.
1.C	Why sequence matters beyond one judgement	Re-issued as E.1.
2.A, 2.B, 2.C	Impression; worked example; soft cues	Re-issued as E.2, E.3 (extended with the soft-versus-hard passage), E.4.
3.A, 3.B	Commutativity literature; Asch	Re-issued as E.5, E.6.

Entry	Subject	Status and what it needs
3.C	Heckman	Superseded by C.15, which carries the identification precedent with P17a applied.
3.D	Bohren-Imas-Rosenberg	Re-issued as E.7.
4.A, 5.A	Section titles and openings	Re-issued as E.8, E.9.
5.B	Propositions ORD and ADJ	Re-issued as E.10 (mathematics unchanged; prose corrected per P10/P11).
6.A, 6.B	Rival mechanisms; two channels	Re-issued as E.11, E.12 (6.B's text pulled in from interior_omega.tex).
B.A, B.B	Bibliography; AI declaration	Re-issued as E.14, E.13.
W.A	Input/output tables	Withdrawn by the author 2026-09-21; kept in notes/empirical_analytics.tex.

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